REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Textile Testing II Lab

A full-screen, syllabus-style digital handbook for Course Code 6068. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6068

COURSE TITLE
Textile Testing II Lab

DEPARTMENT
Textile Technology

TYPE
Lab

SEMESTER
6

CATEGORY
Practical

USE
Study + notes PDF

FORMAT
Big handbook

6068

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map objectives + outcomes	**Module lessons** 4 detailed units
Formula bank calculations	**Practice bank** questions + tasks
Model paper official style	**Answer key** full points

Course objectives and syllabus map

Objectives

- ✓ Perform advanced fibre, yarn and fabric tests
- ✓ Calculate and interpret test results using standard methods
- ✓ Prepare accurate lab reports with conclusion and precautions

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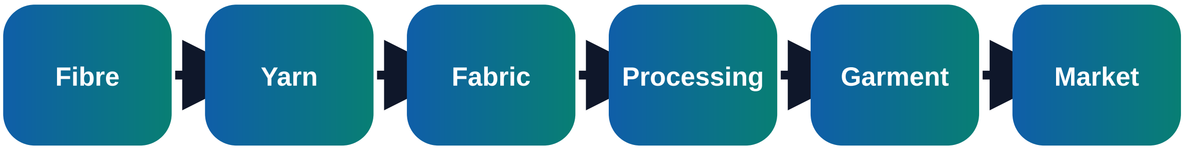
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Lab discipline and sampling	Atmospheric conditions for textile testing; Sampling plan, specimen preparation and conditioning	Short notes, calculations, diagram, checklist, viva answers
M2	Yarn and fibre tests	Yarn count, twist, single yarn strength and lea strength; Evenness, imperfections, hairiness and CSP interpretation	Short notes, calculations, diagram, checklist, viva answers
M3	Fabric mechanical tests	GSM, thickness, tensile strength, tear strength and bursting strength; Abrasion resistance, pilling and crease recovery	Short notes, calculations, diagram, checklist, viva answers
M4	Colour fastness and reporting	Fastness to washing, rubbing, perspiration and light; Grey scale and staining scale assessment	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Lab discipline and sampling

Core explanation

- ✓ Atmospheric conditions for textile testing
- ✓ Sampling plan, specimen preparation and conditioning
- ✓ Instrument calibration, zero error and safe handling
- ✓ Observation table, units, significant figures and report format

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

Yarn and fibre tests

Core explanation

- ✓ Yarn count, twist, single yarn strength and lea strength
- ✓ Evenness, imperfections, hairiness and CSP interpretation
- ✓ Fibre length, fineness, maturity and moisture regain
- ✓ Statistical treatment of test results

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 3

Fabric mechanical tests

Core explanation

- ✓ GSM, thickness, tensile strength, tear strength and bursting strength
- ✓ Abrasion resistance, pilling and crease recovery
- ✓ Drape, air permeability and water repellency
- ✓ Dimensional stability and shrinkage

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

Colour fastness and reporting

Core explanation

- ✓ Fastness to washing, rubbing, perspiration and light
- ✓ Grey scale and staining scale assessment
- ✓ Comparison with specifications and acceptance criteria
- ✓ Final test certificate preparation

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

GSM = Specimen Weight / Area in square metre

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

Moisture Regain % = Moisture Weight / Dry Weight x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

CSP = Count x Lea Strength

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Shrinkage % = Original Length - Final Length / Original Length x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Determine GSM and thickness of fabric
- ✓ Test yarn strength and calculate CV
- ✓ Perform rubbing fastness test
- ✓ Prepare complete lab test certificate

Important keywords

GSM · CSP · CV · Conditioning · Fastness · Abrasion · Pilling · Shrinkage

Short questions

Q1.1	Explain: Atmospheric conditions for textile testing	Answer with definition, process, application and one example.
Q1.2	Explain: Sampling plan, specimen preparation and conditioning	Answer with definition, process, application and one example.
Q1.3	Explain: Instrument calibration, zero error and safe handling	Answer with definition, process, application and one example.
Q2.1	Explain: Yarn count, twist, single yarn strength and lea strength	Answer with definition, process, application and one example.
Q2.2	Explain: Evenness, imperfections, hairiness and CSP interpretation	Answer with definition, process, application and one example.
Q2.3	Explain: Fibre length, fineness, maturity and moisture regain	Answer with definition, process, application and one example.
Q3.1	Explain: GSM, thickness, tensile strength, tear strength and bursting strength	Answer with definition, process, application and one example.
Q3.2	Explain: Abrasion resistance, pilling and crease recovery	Answer with definition, process, application and one example.

Q3.3	Explain: Drape, air permeability and water repellency	Answer with definition, process, application and one example.
Q4.1	Explain: Fastness to washing, rubbing, perspiration and light	Answer with definition, process, application and one example.
Q4.2	Explain: Grey scale and staining scale assessment	Answer with definition, process, application and one example.
Q4.3	Explain: Comparison with specifications and acceptance criteria	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to GSM. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Textile Testing II Lab.
2. List four important keywords: GSM, CSP, CV, Conditioning.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Lab discipline and sampling.
7. Compare two important concepts from Yarn and fibre tests.
8. Explain the practical importance of Fabric mechanical tests in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Textile Testing II Lab in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Textile Testing II Lab.
2. Use keywords: GSM, CSP, CV, Conditioning, Fastness, Abrasion, Pilling, Shrinkage.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Lab discipline and sampling

- Atmospheric conditions for textile testing
- Sampling plan, specimen preparation and conditioning
- Instrument calibration, zero error and safe handling
- Observation table, units, significant figures and report format

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

Yarn and fibre tests

- Yarn count, twist, single yarn strength and lea strength
- Evenness, imperfections, hairiness and CSP interpretation
- Fibre length, fineness, maturity and moisture regain
- Statistical treatment of test results

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Fabric mechanical tests

- GSM, thickness, tensile strength, tear strength and bursting strength
- Abrasion resistance, pilling and crease recovery
- Drape, air permeability and water repellency
- Dimensional stability and shrinkage

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

Colour fastness and reporting

- Fastness to washing, rubbing, perspiration and light
- Grey scale and staining scale assessment
- Comparison with specifications and acceptance criteria
- Final test certificate preparation

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.